(Assignment-1)

ASSIGNMENT - 1
G.I.E.T. UNIVERSITY
GUNUPUR

Pre-requisite of UNIX OS:
Knowledge of basic computer software and hardware and a few features of windows OS.

UNIX is a multi user operating system developed at AT & T BELL laboratory, NEW JERSEY at USA, CALIFORNIA by DENNIS RITCHIE.

Initially it was termed as UNICS as MULTICS (Multiplexed Information and Computing Service), which was also developed using a C programming language.

The UNIX consisting of three components:

1. UNIX SHELL
2. KERNEL and
3. Tools and applications

The UNIX supports various kinds of features:
(1) Multi-user system: When multiple users are accessing a single system.
(2) Multi tasking capability: When system allows to do multiple task concurrently.
(3) Portability: When OS transfer from one to another. (Open source software. Example: Antivirus)
(4) Hierarchical file system: In UNIX the file system hierarchy starts with / as root (/)

Hierarchical file system

Architecture of UNIX

(User1, User2, User3, User4 around shell & kernel diagram)

bin: It stands for binary
dev: It stands for devices
etc: It stands for etcetera
lib: It stands for library
mnt: It stands for mounting (add new drivers)
proc: It stands for process
root: same as windows
temp: It stands for temporary
user: Mostly use lib

5. Security and Permission: Operating system have more secure and more.
Here + means it allows and – means it does not allows.

6. Shell and Scripting Program: The UNIX having two types of scripting program:

1. Bourne shell (sh)
2. Bash shell (Bourne again shell) (BASH)
3. C SHELL (CSH)
4. TENEX C SHELL (TCSH)
5. KORN SHELL (KSH)
6. Z SHELL (ZSH)
7. FISH SHELL (FISH)
8. DASH (DASH)

Basic commands in UNIX:
(How to see a terminal?)
start search for terminal or on desktop right click open terminal.

1. **ls:** Long listing of files and folders in the current working directory.
   → myfile.txt
2. **ls options:**

- ls -l → Long listing of files and folders with full description or information.
- ls -l more → Long listing of files and folders with pipeline and paging.

clear: To clear we use this

→ ls -l
→ drwxrwxrwx → 2 student student 4096 Jan 29 2025 29.01.2025 GIET

3. **man:** It is a helping command which gives the full description of any command.
4. man ls -l : Name
   ls – list directory contents
5. **date:** To see the current date of the terminal.
   → Thursday 04 December 2025 12:36:37 PM IST

(Ctrl + shift + (+)) : Zoom-IN
(Ctrl + shift + (-)) : Zoom-OUT

6. date +%m : Only month to be display.
   → 12
7. date +%b : only to display month name.
   → Dec
8. date +%d : only date to be display
   → 04
9. date +%y : only year is displayed.
   → 25
10. date +%H : Display hour
    → 12
11. date +%T : Display the time
    → 12:45:35
12. date +%M : Display the current minute.
    → 44
13. date +%S : Display the current seconds values.
    → 33

14. cal : (Calendar command)
    Display the current month calendar
15. cal 2025 :
    Display the whole month calendar of 2025

16. cal 12 2025 : Display calendar of specific month i.e 12 of specific year i.e 2025.
17. echo : Welcome to UNIX LAB; one time it will echo welcome to UNIX lab
18. who : It will show the current users who is logged in.
    It will give the full information of the users

→ student :0 2025-12-04 11:40 (:0)

19. who am i : It will show the information of current user who has logged in.
20. uptime : It will show the time till now users have working the terminal.

→ 12:54:31 up 1:14, 1 user, load average : 0.00, 0.00, 0.00

21. uname : It gives the hardware platform information
    → Linux
22. hostname : It will display the system name.
    → Lab01 - PC6
23. bc (Batch Calculator) : It is not only used for arithmetic operation but also used in loop.

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How to kill any command? → Ctrl + C (Explicitly kill a command)

2. quit (just come out from the command)
3. Esc + i

Cat :- cat command is used for various purpose
like : (i) creating a file
(ii) reading the content of a file

(i) Creating a file :-
cat > <filename.txt>
add data
ctrl + z & save the file

creating a directory : mkdir <directory name>
changing the directory : cd <directory name>

Reading the content of a file :-
cat <file name>

cat for combine file data to another file :-
cat <existing file> > <new file>

grep : Grep is a command that allows to search a particular word or pattern from an existing file.
The grep will give the searched data in different colour.
If the pattern don't match then the output is null.

remove a file :-
rm filename.ext

ex:- rm hello.txt (or) rm hello.*

copy a file :-
cp <filename>

cp command allows to copy a file or multiple files from one place to another place with same content and same or different name.

cp <sourcefile> <destination file>

head : It is a command to read the top lines of a file.

tail : It is a command to read the bottom lines of a file.

Creating a file with different approach :-
touch, cat, vi, gedit are some approach to creating a file.

touch : It can create an empty file if the file is not available.

touch myfile.txt
ls my*
mv file1 file2

file : file command is used to know the type of a file
Ex :- file demo.txt → ASCII text

vi : Vi stands for visual interface editor

The vi consists of 3 modes :-
(i) visual mode / read mode
(ii) insert mode / writing data mode
(iii) append mode / updating mode

The first mode is visual mode, second mode is insert and last mode is append

Syntax : vi <file.txt>

To write any data (i)
To append any data (ii)

Saving data : esc + shift + : (:) → wq

zip and unzip

zip command is used to compress one or multiple files into a zip archive directory and to send all data from a compressed archive directory.

Compress a file :-
Syntax : zip myzip.zip <file name>

zip myzip.zip demo.txt
adding : demo.txt (deflated 43%)

unzip myzip.zip

Exclude files while zipping :-
zip -x myzip.zip *.exe

Extract the zip compressed are directories :-
unzip myzip.zip

chmod : chmod is called as changing of mode of a file or directories.

In Linux any file contains 3 type of behaviour/properties and can have multiple with 3 user.

The behaviour are : r = read
w = write
x = executable

and the manipulatory are : u = owner
g = group
o = other